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Cells

Eukaryotic vs Prokaryotic

Bacterial cells vs Vertebrate cells



• What Are the Similarities?

- **DNA:** Genetic coding that determines all the characteristics of living things. Both can evolve and adapt.
- **Cell (or plasma) membrane:** Outer layer that separates the cell from the surrounding environment and acts as a selective barrier for incoming and outgoing materials.
 - **Cytoplasm:** Jelly-like fluid within a cell that is composed primarily of water, salts and proteins.
 - **Ribosomes:** Nucleic acid organelles that make proteins.

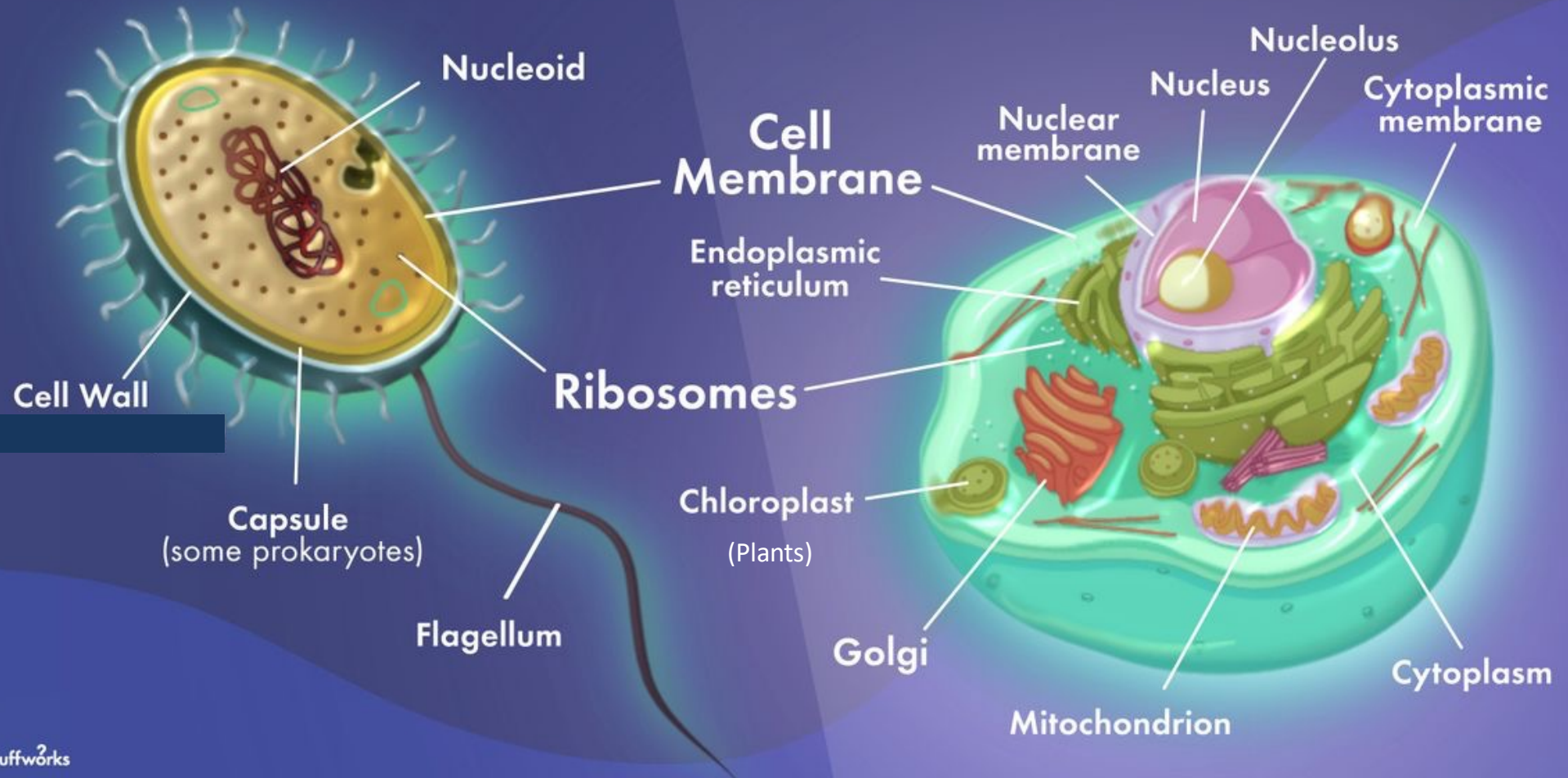
What are the Differences?

- Prokaryotic cells have **no nucleus**; rather, they have a membraneless nucleoid region of the cell interior that holds free-floating DNA (usually a single ring of DNA) and they have only one copy of each gene.
- Prokaryotic cells have **no well-characterized organelles**, although they can still carry out most of the same functions.
- Prokaryotic cells reproduce by **binary fission**. The cell duplicates its DNA, doubles its cytoplasm and divides in two. Rapid and simple.
- Most prokaryotic cells have a **rigid cell wall** that surrounds the plasma membrane and gives shape to the organism.
- Eukaryotic cells have **a nucleus** surrounded by a nuclear envelope, a lipid bilayer membrane. The nucleus holds the eukaryotic cell's DNA, which has two copies of each gene, one from each parent.
- Eukaryotic cells have **membrane-bound organelles**, including mitochondria, rough and smooth endoplasmic reticulum, golgi complex (and in the case of plant cells, chloroplasts (conduct photosynthesis)).
- Eukaryotic cells reproduce by **mitosis/meiosis**
- Among eukaryotes, vertebrates have **no cell wall**, but plants do.

The difference between the prokaryotic cell wall and the eukaryotic cell membrane is one of the key differences that allows certain chemicals to destroy cell walls without harming cell membranes. This is the basis for several classes of antibiotics used to treat bacterial infections.

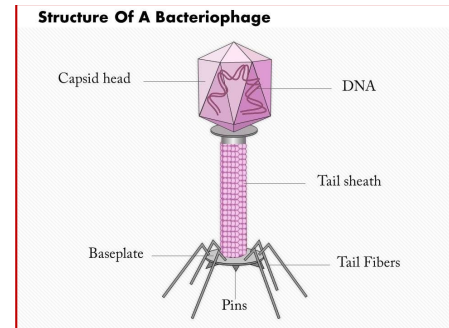
Prokaryotes

Eukaryotes

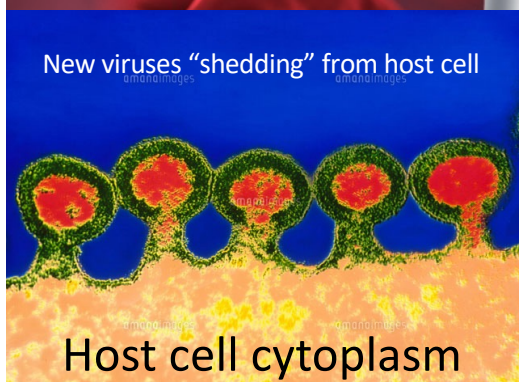




Is a Virus a Cell?

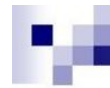


- No, a virus is not a cell. A virus is a non-living infectious agent that can replicate only inside the living cells of another organism-not on its own.
- They have either DNA or RNA as their genetic material, enclosed in a protein shell called a **capsid**. If they have RNA (e.g. HIV) they must first convert it to DNA before replicating. For this purpose, RNA viruses have a special enzyme called reverse transcriptase.
- Some viruses have an additional coating around the capsid called an **envelope**.
- Their surface has specialized proteins (red spikes in upper photo) that allow it to attach to and infect host cells. Once inside, they take control of the protein machinery of the cell, make more of themselves, and the progeny escape the host cell by “shedding” (see colorized EM bottom left), sometimes killing it along the way and sometimes not.
- Some viruses are specialized to infect only bacteria. These are called bacteriophages (or just phage viruses). They have a specialized structure shown in upper right.

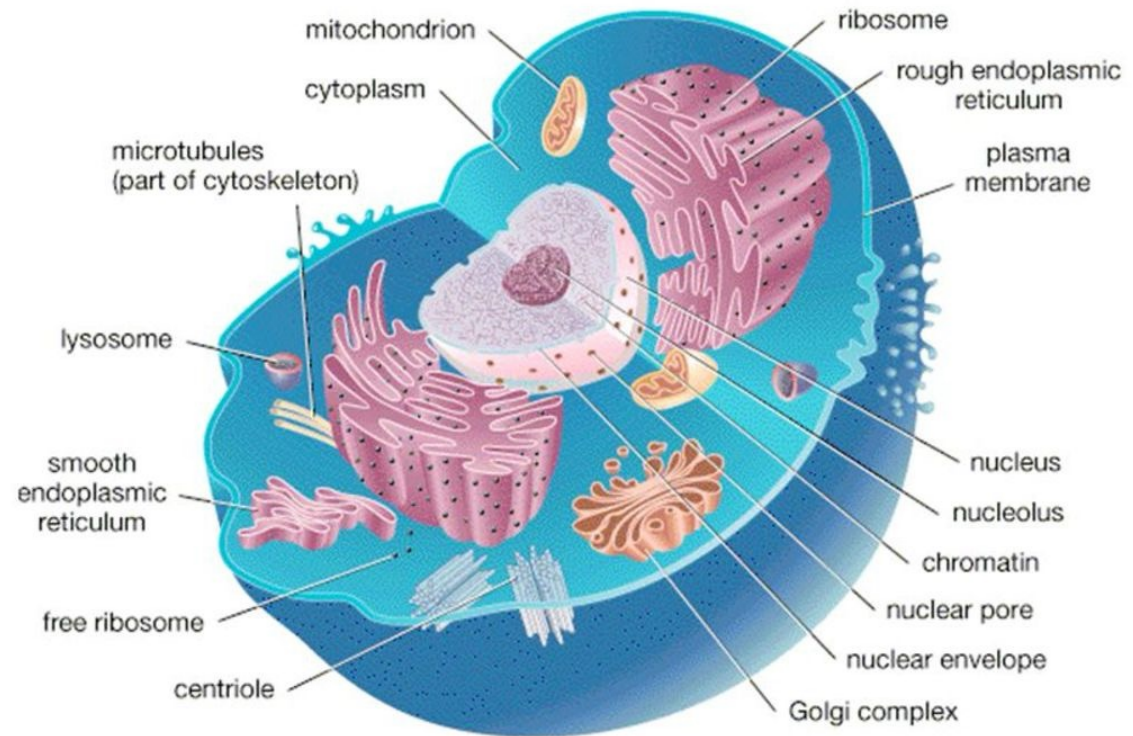


Eukaryotic Cell

- Cell Membrane
- Nucleus
 - Chromatin
 - Nucleolus
 - Nuclear envelope
 - Nuclear pores
- Cytoplasm
 - Cytosol
 - Endoplasmic reticulum
 - Smooth or rough
 - Ribosomes
 - Golgi apparatus
 - Mitochondria
 - Centriole
 - Lysosome
 - Cytoskeleton

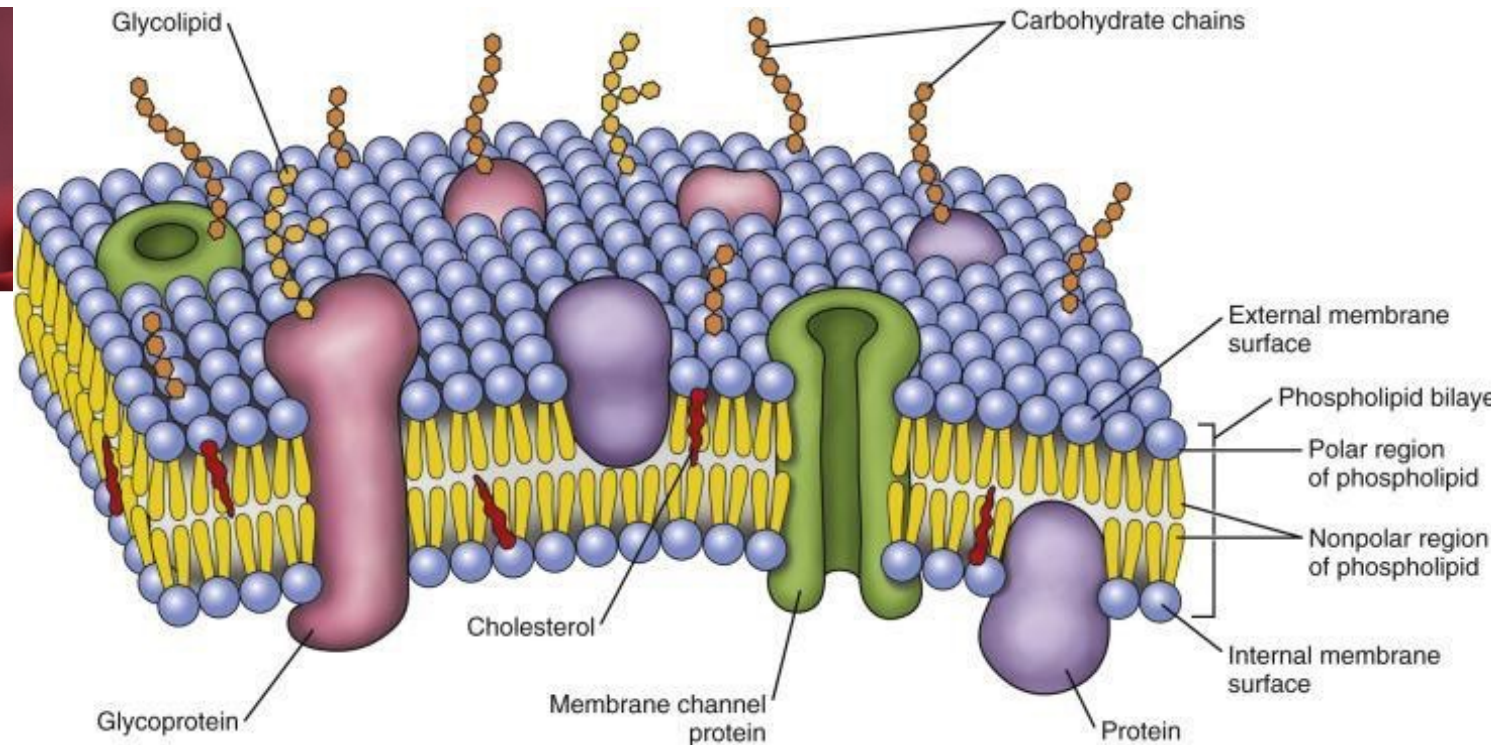


Eukaryotic Cell (protist, animal)



Cell Membrane

- **Cell Membrane**
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The cell membrane is a lipid bilayer composed mostly of two layers of phospholipids. Each phospholipid molecule has a hydrophilic head (aka polar) and two hydrophobic (aka nonpolar) tails. The hydrophilic head of one layer faces the inside of the cell and the hydrophilic head of the other layer faces the outside. The hydrophobic tails of both layers face each other and are buried. By electron microscopy this looks like a set of railroad tracks (EM photo of two adjacent membranes at right).



Floating in this lipid bilayer are an array of specialized proteins, serving as receptors, ports, channels, pumps or cell adhesion and communication structures.

Cell Nucleus

- Cell Membrane

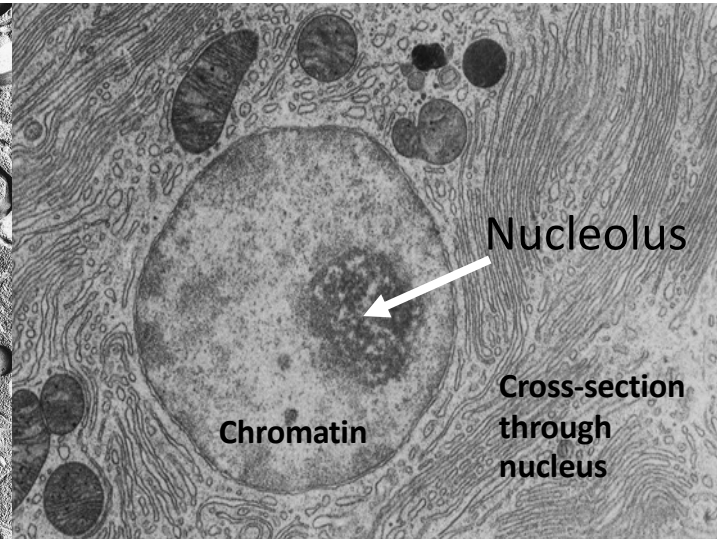
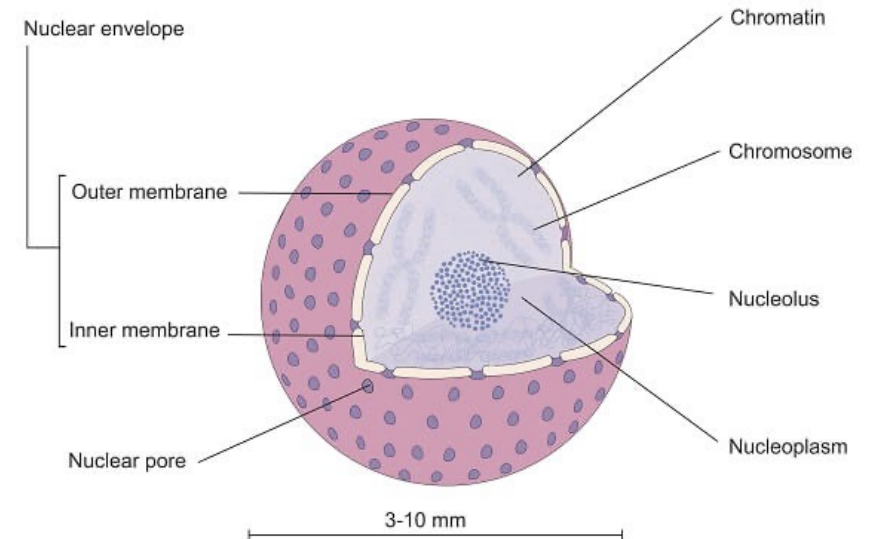
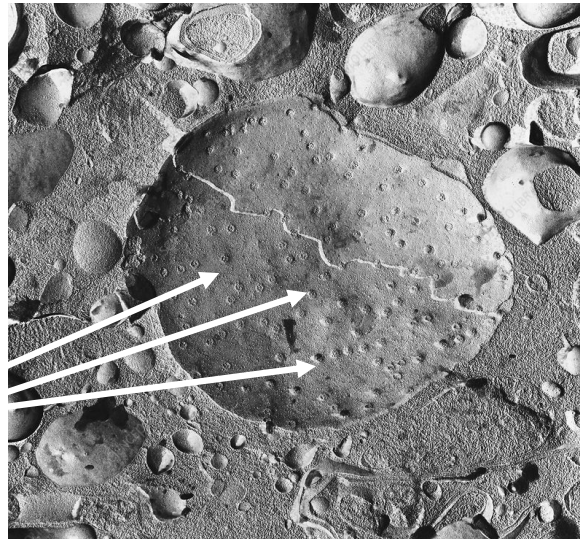
- Nucleus

- Chromatin – the tightly coiled chromosomes of DNA
- Nucleolus – contains the genes that encode rRNA
- Nuclear envelope- double membrane surrounding nucleus
- Nuclear pores- passageways for moving molecules into and out of nucleus

- Cytoplasm

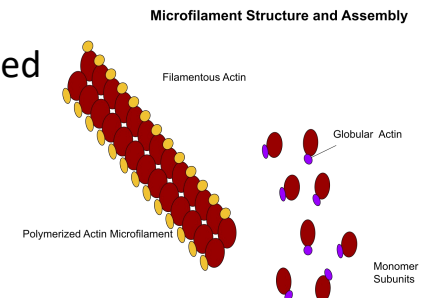
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Surface view of nucleus showing pores in nuclear envelope.

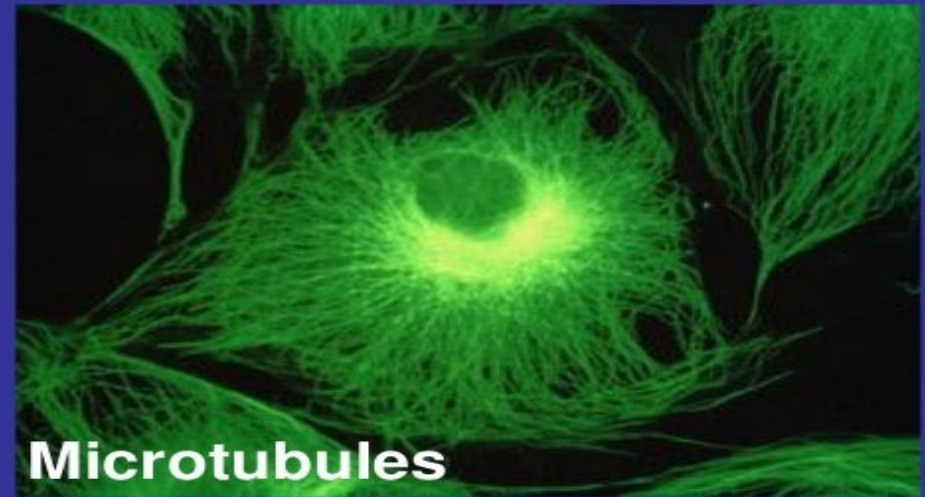


Cytosol/Cytoskeleton

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- The **Cytoplasm** includes the cytosol, the cytoskeleton and all the organelles. The **Cytosol** is the liquid inside the cell membrane that surrounds the organelles and cytoskeleton. The cytosol is a complex solution where most of the cellular biochemistry takes place.
- The cytoskeleton is a network of intracellular protein fibers that provides structural shape and stiffness, anchors organelles, and facilitates intra- and extracellular movement of cell materials. Eukaryotic cells contain 3 main kinds of cytoskeletal filaments: **microfilaments, microtubules and intermediate filaments**. In neural cells, intermediate filaments are called neurofilaments.
- **Microfilaments** are mostly filamentous actin (F-actin) formed by polymerization of globular (G-actin) molecules.
- **Microtubules** are thin tubes formed by proteins called tubulins and are involved in intracellular transport, cell division and the structural elements of cilia.
- **Intermediate Filaments** – filaments that rearrange to modify cell shape and rigidity, including proteins in families such as desmin, glial fibrillary acidic protein (GFAP), vimentin and syncoilin.



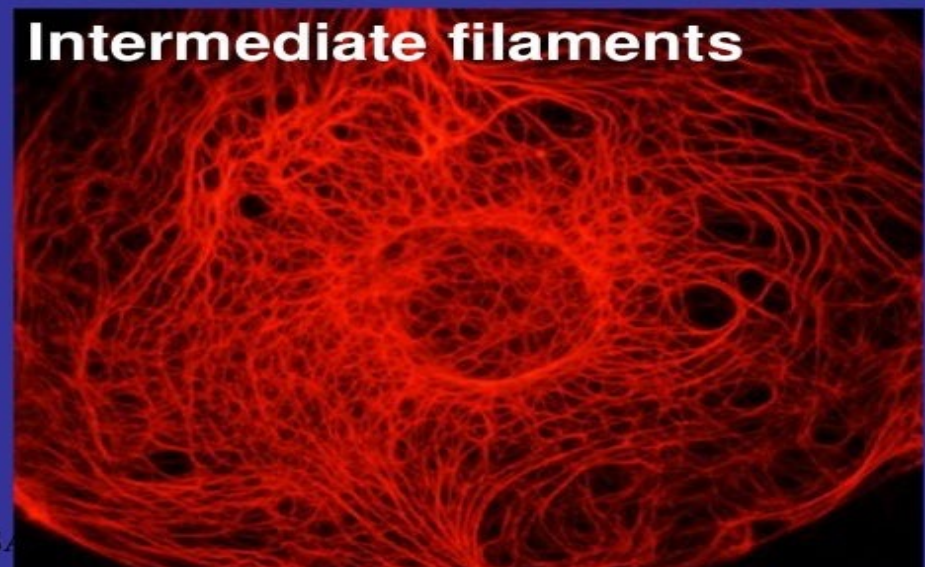
Cells have three
cytoskeletal elements



Microtubules



Microfilaments



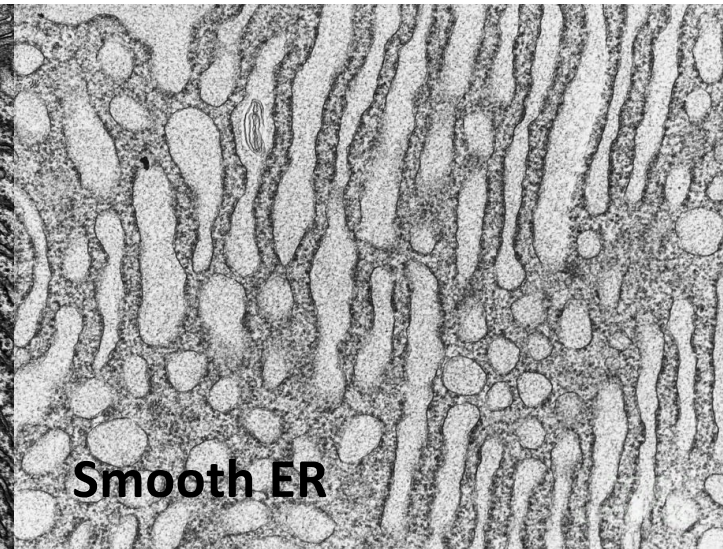
Intermediate filaments

Endoplasmic Reticulum

- The **endoplasmic reticulum (ER)** forms an interconnected network of flattened, membrane-enclosed sacs known as cisternae. They serve as the sites of cell product synthesis. This organelle is composed of two subunits – **rough endoplasmic reticulum (RER)**, and **smooth endoplasmic reticulum (SER)**. The RER plays a central role in protein synthesis and protein folding. The SER functions primarily in lipid synthesis. In muscle cells, there is smooth ER-like structure called the sarcoplasmic reticulum.
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The many black dots that give the RER its "rough" appearance, are ribosomes



Smooth ER

Making protein in Rough ER: Transcription vs Translation



Transcription

- To Transcribe is to copy text within the same language.
 - In protein synthesis, the DNA **nucleotide** code is transcribed to a messenger RNA **nucleotide** code.
- Although we are going from DNA to RNA, we remain within the nucleotide code language of sequences of purines and pyrimidines (guanine, cytosine, adenine and thymidine), so it is transcription.

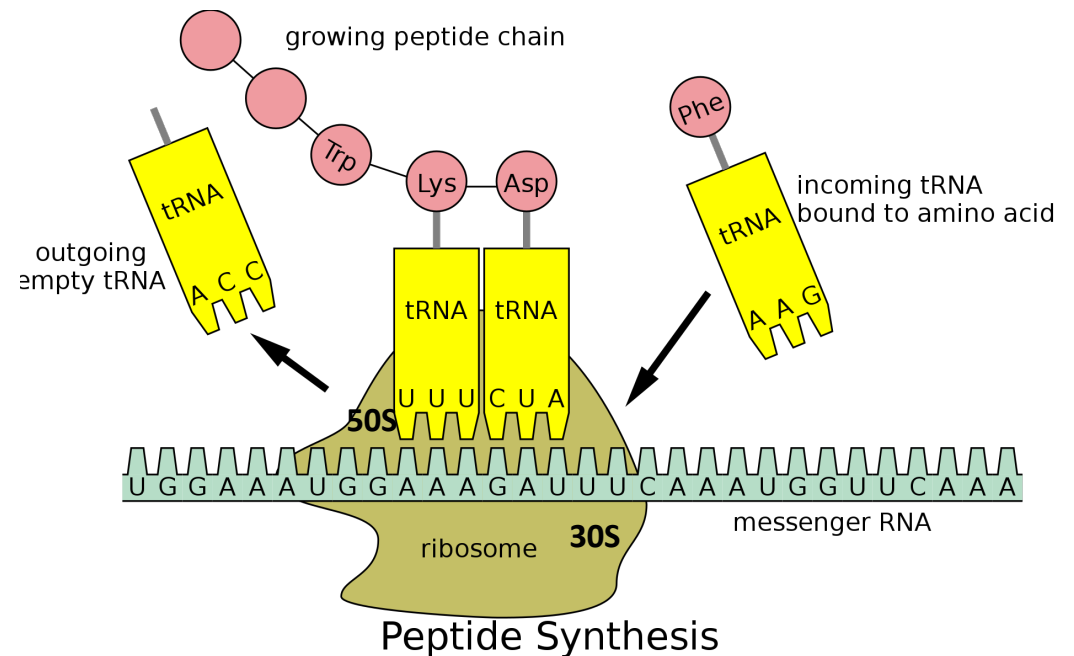
Translation

- To Translate is to copy text, but also turn it into a new language.
 - In protein synthesis, the mRNA language of **nucleotide** codes (sequences of GCAT) is translated into the code of proteins, which is sequences of **amino acids**
- In this step, which occurs in the rough ER, we have changed languages from codes made of sequences of purine and pyrimidines to sequences of amino acids, so this is a translation

Transfer RNA, Messenger RNA and Ribosomes



- mRNA, with the transcribed code from DNA, leaves the nucleus for the cytoplasm. Molecules of transfer RNA read the sequence of the mRNA code, determine which amino acid corresponds to that code sequence, picks up the amino acid and delivers it to the ribosome, which assembles the amino acids, in the proper sequence, to create the desired peptide sequence for the desired protein.
- Ribosomes: A ribosome is made from RNAs and proteins. Each ribosome is composed of small (30S) and large (50S) component, called subunits, which are bound to each other across the mRNA
- The 30S subunit functions mostly in decoding and is also bound to the mRNA
- The 50S subunit serves mainly a catalytic function assembling the amino acids into peptides and thus is more associated with the tRNA.



Golgi Apparatus

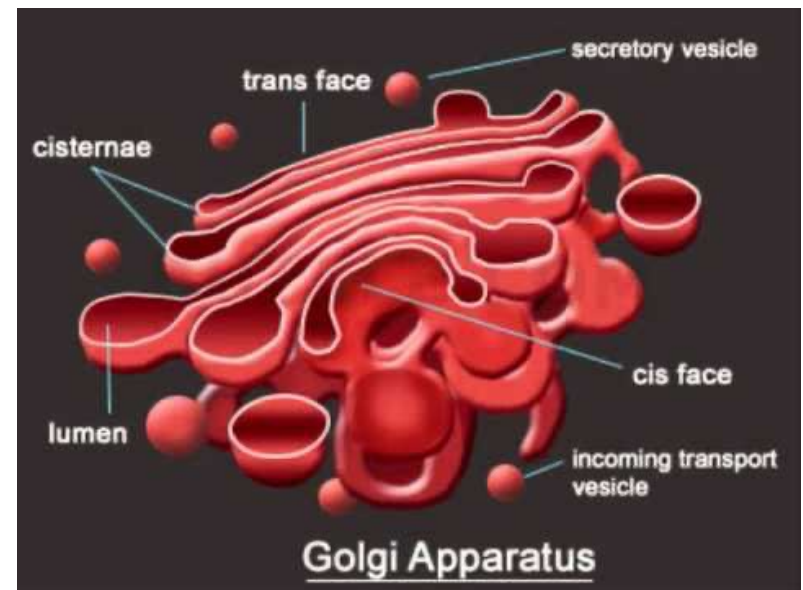
Stacks of Golgi

Nucleus

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- The Golgi apparatus is the packaging and shipping center of the cell, responsible for transporting, modifying, and packaging proteins and lipids into vesicles, usually for secretion. It is always positioned right next to the nucleus.

The cisternae of the Golgi are described as having two faces –cis and trans. The cis face receives products synthesized by the ER and releases them in secretory vesicles from the trans face. These vesicles move to the cell membrane where they fuse with the membrane and release their product.

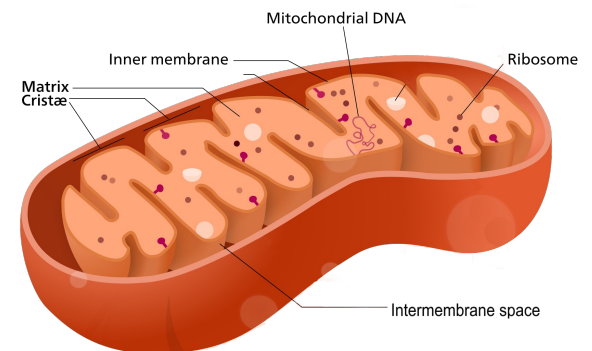


Mitochondria

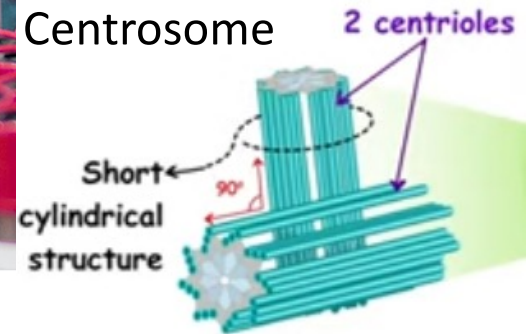
Singular = mitochondrion

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- Mitochondria have a double membrane structure (an outer membrane and a convoluted inner membrane forming chambers called cristae). They use aerobic respiration and an electron transport chain to generate adenosine triphosphate (ATP), which is used throughout the cell as a source of chemical energy. They also play a role in calcium regulation and storage. In one cell type (brown fat), mitochondria generate heat instead of ATP. Importantly, mitochondria have their own DNA, which is invariably inherited from the mother and not the father because eggs have loads of mitochondria and sperm almost none. Size and number of mitochondria vary widely in different cell types, usually based upon energy demand. The outer membrane contains enzymes involved in other activities such as the elongation of fatty acids and oxidation of adrenalin.

Black dots inside mitochondrion are calcium. Note they are larger than surrounding ribosomes on ER

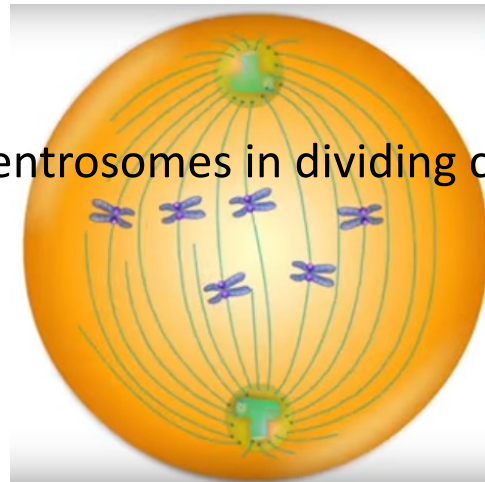


Centriole

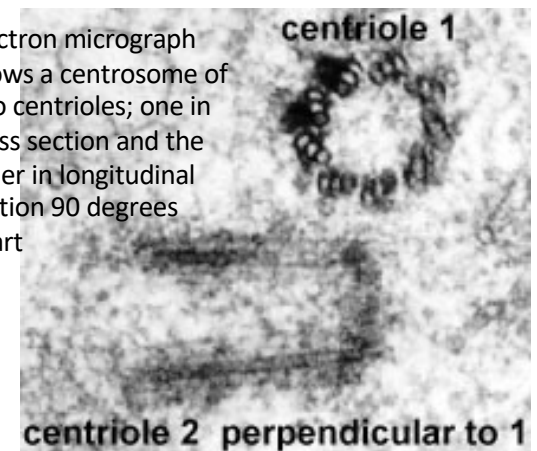


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- Centrioles are short tubular arrays of microtubules, usually found in pairs positioned 90 degrees apart. Together, these form a centrosome. They are involved primarily in cell division. For cell division to occur, the centrosome duplicates itself and the 2 centrosomes move to opposite poles of the cell. From each, a microtubular array forms that attaches to the chromosomes and each pulls one set to themselves, allowing for the cell to divide in the middle. Chemically interfering with centriole function and microtubule formation is one of the main ways used to prevent cell division in cancer.

Centrosomes in dividing cell



Electron micrograph shows a centrosome of two centrioles; one in cross section and the other in longitudinal section 90 degrees apart



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- There are primary and secondary lysosomes. **Primary lysosomes** are membrane-bound vesicles that contain a very acidic environment, including enzymes that operate at low pH like we see in the stomach.

They are used for the digestion of macromolecules, old cell components, and microorganisms. They are formed by the Golgi apparatus. These primary vesicles fuse with larger, membrane-bound vesicles, called **phagosomes**, containing material arising from three main processes: endocytosis, autophagocytosis, and phagocytosis. Once the primary lysosome and phagosome fuse, the resulting structure is called a **secondary lysosome** and it is in the secondary lysosome that the material in the phagosome is digested. The secondary lysosome can export the digested material from the cell. Lysosomal storage diseases are genetically inherited deficiencies that leave material undegraded and so it builds up in various tissues and organs. One of these is Hurler's syndrome in which mucopolysaccharides will build up in many tissues, including the cornea. You could be the one who makes the diagnosis of this disease.

